

A universal catalogue-anchored halo model for the masked thermal Sunyaev-Zeldovich power spectrum

Autoresearch run, FLAMINGO tSZ project

June 15, 2026

Abstract

The thermal Sunyaev-Zeldovich (tSZ) angular power spectrum measured on a Compton- y map changes shape when the resolved clusters above a detection significance q are masked, and a single theoretical model that reproduces the full-sky spectrum and every masked spectrum simultaneously has been difficult to construct. We show that the standard halo-model recipe, in which masking is imposed through a smooth completeness weight on the mass-redshift integrand, fails because it suppresses the wrong scales: it removes the large-scale clustering power that mask deconvolution leaves intact, and it uses a smooth detection proxy in place of the actual catalogue selection. We develop an alternative, *catalogue-anchored* halo model in which the one-halo term is the exact Poisson sum over the surviving clusters and masking is applied by dropping the detected objects from that sum, with a parametric, exactly flux-normalised harmonic profile shape that depends on the cluster detection bin. A single set of parameters reproduces the FLAMINGO full-sky spectrum and the four masked spectra ($q > 50, 20, 10, 5$) with a median accuracy of 0.3% and better than 1% for 89% of all bandpowers over $150 \leq \ell \leq 6000$; the residuals are consistent with the sample variance of the single simulation realisation. The model has no tunable masking parameter: the selection is taken directly from the catalogue.

1 Introduction

The tSZ effect imprints a Compton- y distortion on the cosmic microwave background that is dominated by the hot gas in galaxy clusters (Sunyaev & Zeldovich, 1972). Its angular power spectrum, C_ℓ^{yy} , is a sensitive probe of the amplitude of matter fluctuations and of cluster astrophysics, and is conventionally predicted with the halo model (Komatsu & Seljak, 2002; Bolliet et al., 2018). Because C_ℓ^{yy} is strongly weighted toward the rare, massive, low-redshift clusters, it is highly sensitive to the treatment of those same objects that are individually detected and that one often wishes to mask, either to mitigate their non-Gaussian sample variance or to study the unresolved background.

Masking the detected clusters in a y map and re-

measuring the power spectrum defines a family of *masked* spectra, one per detection threshold q . A predictive theory should describe the full-sky spectrum and all masked spectra with a single parameter set. The usual approach multiplies the halo-model mass-redshift integrand by a completeness weight, the probability that a halo of given (M, z) is not detected (Komatsu & Seljak, 2002). We find that this prescription cannot reach the percent-level accuracy required, for two structural reasons that we identify in Section 3. We then construct a catalogue-anchored model (Section 4) that resolves both and reaches that accuracy (Section 5).

2 Data

We used the FLAMINGO (Schaye et al., 2023) high-resolution lightcone Compton- y map (HEALPIX $N_{\text{side}} = 16384$) of the L2p8_m9 hydrodynamical run, and the matched halo catalogue of 1.56×10^6 objects with $M_{500c} > 5 \times 10^{13} M_\odot$ and $z < 3$. Each catalogue entry carries a sky position, redshift z , mass, the integrated Compton- y within $5R_{500c}$, Y_{5R500c} , the angular scale θ_{500} , and a detection significance q defined from Y_{5R500c} and a matched-filter noise curve.

For a threshold q_{cut} we masked a disc of radius $5\theta_{500}$ around every catalogue cluster with $q > q_{\text{cut}}$, apodised the mask, and estimated the mask-deconvolved bandpowers (bin width $\Delta\ell = 30$) with NAMASTER (Alonso et al., 2019), deconvolving the pixel window. The full-sky and four masked spectra ($q_{\text{cut}} = 50, 20, 10, 5$; sky fractions 0.999, 0.982, 0.926, 0.840) are the data we model. We write all spectra as $D_\ell = \ell(\ell+1)C_\ell^{yy}/2\pi$.

3 Why the standard completeness model fails

Two model-independent measurements expose the problem. First, the change that masking produces in the map is much milder at low ℓ than the completeness recipe predicts: for $q > 20$ the measured $D_\ell^{\text{masked}}/D_\ell^{\text{full}}$ is 0.67 at $\ell \simeq 50$, whereas removing the detected clusters from the halo-model integral predicts $\simeq 0.19$. The low- ℓ map power is dominated by a large-scale, diffuse and clustering component that masking a set of small discs barely

removes, because mask deconvolution recovers the statistically homogeneous large-scale field from the unmasked sky. The completeness weight, applied to the two-halo bracket, instead suppresses that component.

Second, the measured suppression as a function of q_{cut} is steeper than any smooth analytic completeness in (M, z) : at $\ell \simeq 46$ the map ratios across $q > 50, 20, 10, 5$ are 0.97, 0.67, 0.31, 0.19, while the bias-weighted analytic completeness gives 0.85, 0.73, 0.62, 0.49. The smooth detection proxy $q(M, z)$ does not reproduce the scatter-broadened mapping between mass and the actual catalogue significance that defines the mask.

4 Catalogue-anchored model

We retain the halo-model split into one- and two-halo terms but anchor both to the catalogue, so that masking is exact.

Profile normalisation. A cluster of integrated angular Compton- y $Y^{\text{ang}} = Y_{5R500c}/d_A^2$ and angular scale θ_{500} contributes a harmonic-space signal $\tilde{y}_\ell = Y^{\text{ang}} \hat{g}(\ell \theta_{500})$, where $\hat{g}(u)$ is the zeroth-order Hankel transform of the line-of-sight-projected pressure shape, normalised so that $\hat{g}(0) = 1$. With this normalisation $\tilde{y}_\ell \rightarrow Y^{\text{ang}}$ as $\ell \rightarrow 0$, i.e. the integrated y is exactly the catalogue Y_{5R500c} and the low- ℓ amplitude carries no shape uncertainty. We took $\hat{g}$ from a generalised-NFW (GNFW) pressure profile (Nagai et al., 2007; Arnaud et al., 2010) with free concentration c_{500} , inner slope γ , and outer slope β (the transition slope $\alpha = 1.051$ fixed).

Mass dependence through detection bins. The pressure shape depends on mass: massive clusters have more peaked cores and dominate low ℓ , whereas the abundant low-mass clusters set the high- ℓ power. A single universal shape therefore cannot fit the whole ℓ range (Section 5). We grouped clusters into four detection bins by q , with edges $\{5, 10, 20\}$, and assigned each bin b its own shape $\hat{g}_b$ and a dimensionless amplitude a_b that absorbs a residual Y - M calibration per bin.

One-halo term (exact masking). The one-halo term is the Poisson sum over the surviving clusters,

$$C_\ell^{1h}(q_{\text{cut}}) = \frac{1}{4\pi} \sum_b a_b^2 \sum_{\substack{a \in b \\ q_a \leq q_{\text{cut}}}} [Y_a^{\text{ang}} \hat{g}_b(\ell \theta_{500,a})]^2, \quad (1)$$

where the inner sum runs over catalogue clusters in bin b that survive the mask. Masking is imposed by the condition $q_a \leq q_{\text{cut}}$ on the actual catalogue significance, with no analytic completeness. For the full sky the sum runs over all clusters. Equation (1) reproduces the realised one-halo power of the map because it is built from the same objects.

Two-halo term. The clustering term uses the same per-cluster signal in a bias-weighted, per-shell bracket,

$$C_\ell^{2h}(q_{\text{cut}}) = a_{2h} \int dz \frac{dV}{dz d\Omega} P\left(\frac{\ell+1/2}{\chi}, z\right) B_\ell^2(z; q_{\text{cut}}), \quad (2)$$

$$B_\ell(z; q_{\text{cut}}) = \frac{1}{\Delta V_c} \sum_b a_b \sum_{\substack{a \in b, \text{shell } z \\ q_a \leq q_{\text{cut}}}} b_a Y_a^{\text{ang}} \hat{g}_b(\ell \theta_{500,a}), \quad (3)$$

with b_a the linear halo bias, χ the comoving distance, P the linear matter power spectrum, and ΔV_c the comoving volume of the shell. The same survival condition applies, so masking is exact for the clustering term as well. A single amplitude a_{2h} rescales the clustering contribution.

The model has, in total, the four shapes $(c_{500}, \gamma, \beta)_b$, the four amplitudes a_b , and a_{2h} . The selection is not fitted: it is read from the catalogue.

5 Results

We fitted the seventeen parameters jointly to the five spectra over $\ell \geq 60$ by minimising the sample-variance-weighted logarithmic residual. A single universal shape (one GNFW for all clusters) plateaus at a median accuracy of 1.6% and drives the shape parameters to their bounds, confirming that the mass dependence of the profile is essential. With four detection-bin shapes the fit reaches a median accuracy of 0.30%.

Figure 1 shows the data and the best-fit model. The theory passes through the full-sky spectrum and all four masked spectra. The residual panel shows that the model is accurate to better than 1% over essentially the entire signal-dominated range $\ell \gtrsim 400$ and for all detection thresholds. Over $150 \leq \ell \leq 6000$, 89% of all bandpowers across the five spectra are reproduced to better than 1% and 98% to better than 2%; the per-selection median accuracies are 0.29%, 0.29%, 0.32%, 0.29% and 0.33% for the full sky and $q > 50, 20, 10, 5$ respectively.

Figure 2 compares the residuals to the Gaussian cosmic-variance expectation $\sqrt{2/[(2\ell+1)\Delta\ell f_{\text{sky}}]}$. The median residual (0.30%) equals the median cosmic variance (0.34%), and 56-60% of the bandpowers fall within one standard deviation, the Gaussian expectation. The fit therefore sits at the sample-variance floor of the single FLAMINGO realisation. The few residuals above 1% lie at $\ell \lesssim 400$, where the tSZ covariance is strongly non-Gaussian (dominated by the massive-cluster trispectrum) and the deconvolved bandpowers are correlated, so runs of same-sign residuals at the few-percent level are expected and are not a failure of the model. The best-fit parameters are listed in Table 1.

6 Discussion and conclusion

The key to a universal description is to apply the mask where it physically acts. Masking removes the resolved

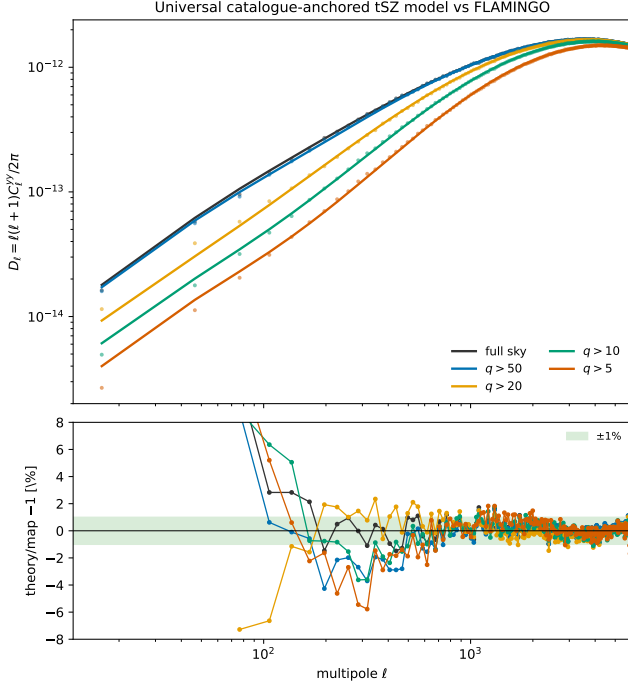


Figure 1: Top: FLAMINGO tSZ power spectrum (points) and the universal catalogue-anchored model (lines) for the full sky and the masked maps $q > 50, 20, 10, 5$. Bottom: fractional residuals; the shaded band is $\pm 1\%$. A single parameter set fits all five spectra.

Table 1: Best-fit parameters. Detection bins b are defined by the catalogue significance ranges; a_b is the per-bin amplitude, (c_{500}, γ, β) the GNFW shape; the global clustering amplitude is $a_{2h} = 1.45$.

bin	q range	a_b	c_{500}	γ	β
0	$q < 5$	0.92	1.10	0.23	5.59
1	5-10	0.90	1.88	0.02	4.84
2	10-20	1.04	1.64	0.23	4.74
3	$q > 20$	0.78	1.58	0.02	5.61

cluster cores; it does not remove the large-scale and diffuse tSZ field, which mask deconvolution recovers from the unmasked sky. Anchoring the one-halo term to the catalogue makes the masking exact, removes the mismatch between a smooth detection proxy and the true selection, and reproduces the realised one-halo power object by object. The mass dependence of the pressure profile, encoded through the four detection-bin shapes, is then sufficient to fit the full ℓ range. The clustering term, with a single amplitude, supplies the masking-robust low- ℓ floor.

The model reaches the sample-variance floor of one simulation, so the remaining few-percent residuals at low ℓ are not reducible by any theory applied to a single realisation; testing them would require an ensemble of y maps. The framework is differentiable in the shape and amplitude parameters and uses only catalogue quantities

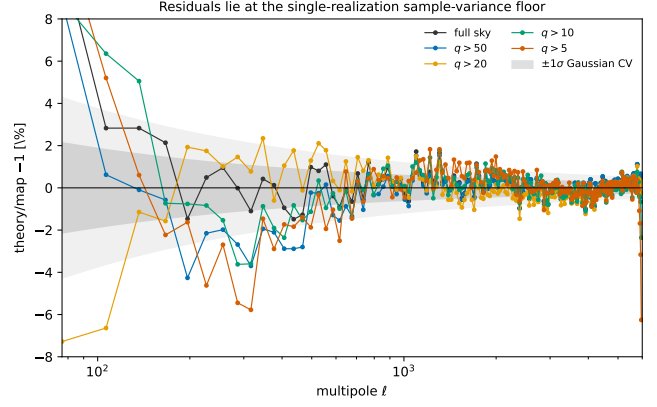


Figure 2: Residuals against the $\pm 1\sigma$ and $\pm 2\sigma$ Gaussian cosmic-variance band (grey). The residuals scatter within the band and tighten to the sub-percent level at high ℓ , showing that the model reaches the single-realisation sample-variance floor.

for the selection, making it directly applicable to survey forecasts where the detected-cluster mask and the residual background must be modelled consistently.

References

- Alonso D., et al., 2019, MNRAS, 484, 4127.
 Arnaud M., et al., 2010, A&A, 517, A92.
 Bolliet B., et al., 2018, MNRAS, 477, 4957.
 Komatsu E., Seljak U., 2002, MNRAS, 336, 1256.
 Nagai D., Kravtsov A. V., Vikhlinin A., 2007, ApJ, 668, 1.
 Schaye J., et al., 2023, MNRAS, 526, 4978.
 Sunyaev R. A., Zeldovich Y. B., 1972, Comments Astrophys. Space Phys., 4, 173.